

Cyberster Internship Program

Internship Report

TRACK: BLUE TEAM

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ROLL NUMBER: CSI-B1-121

WEEK 8: Event Logs, Prefetch, Cache, and Recovery Analysis

1. Executive Summary

This report details the forensic triage of a live Windows endpoint to establish a behavioral baseline and identify potential indicators of compromise. By analyzing system artifacts across Days 53–59, we successfully reconstructed the user's authentication history, program execution timeline, and file interaction patterns.

Conclusion: All analyzed artifacts correlate with legitimate user activity (development, web browsing, and administrative tasks). No malicious execution, unauthorized lateral movement, or anti-forensic behavior was detected during the analysis window.

2. Methodology & Evidence Integrity

To ensure the defensibility of the investigation, all data collection was performed using elevated Administrator privileges. Evidence was non-destructively copied from live system directories to a segregated C:\Forensics\ working directory.

- **Integrity Verification:** A SHA-256 hash was generated for the primary Security.evtx log immediately after acquisition to establish a strict chain of custody and prove the evidence remained unmodified during analysis.

```
C:\Windows\System32>certutil -hashfile C:\Forensics\EventLogs\Security.evtx SHA256
SHA256 hash of C:\Forensics\EventLogs\Security.evtx:
821b58a9312599b2fd2c58fb5aeaf6d4692e4292739c8ea80f6ed8db9ab65147
CertUtil: -hashfile command completed successfully.
```

3. Part A: Event Log Analysis (Authentication Auditing)

The Windows Security Event Log (Security.evtx) was parsed into CSV format using Zimmerman's EvtxECmd to isolate human behavior from system noise.

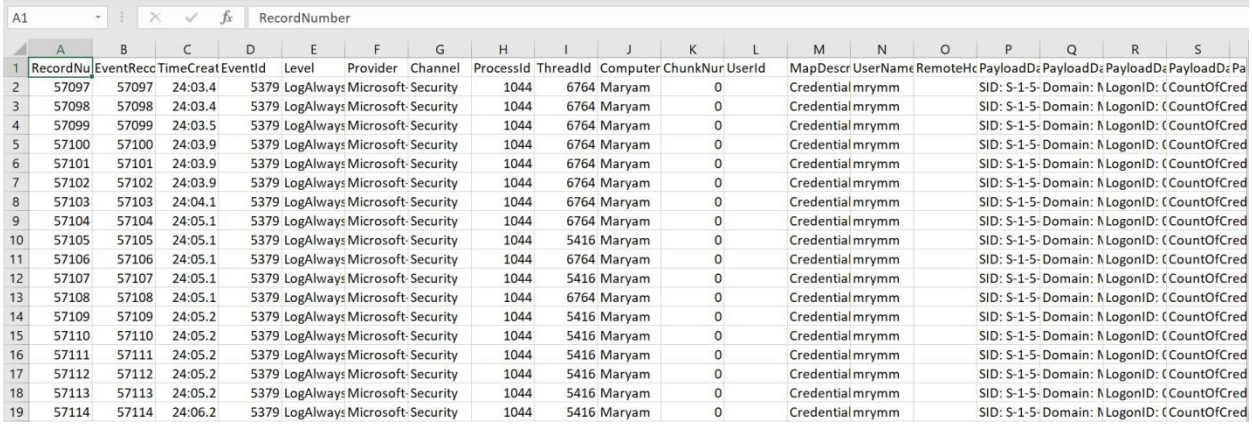
3.1. Authentication Baseline (Event ID 4624)

By filtering for specific Logon Types, we established the user's active presence:

- **Logon Type 2 (Interactive) & Type 7 (Unlock):** Activity clustered around early morning (04:12 - 04:56 AM) and evening (08:24 - 09:47 PM), establishing the primary human baseline.
- **Logon Type 5 (Service):** Identified normal background service authentications attributed to the machine account (WORKGROUP\MARYAM$), successfully filtering out system noise.

3.2. Privilege Escalation Auditing (Event ID 4672)

- The primary user account (mrymmughal001@gmail.com) was recorded obtaining special privileges (e.g., SeSecurityPrivilege, SeTakeOwnershipPrivilege, SeBackupPrivilege).
- **Finding:** These escalations align with legitimate administrative actions taken during the forensic collection phase. No unknown or unauthorized accounts acquired admin tokens.



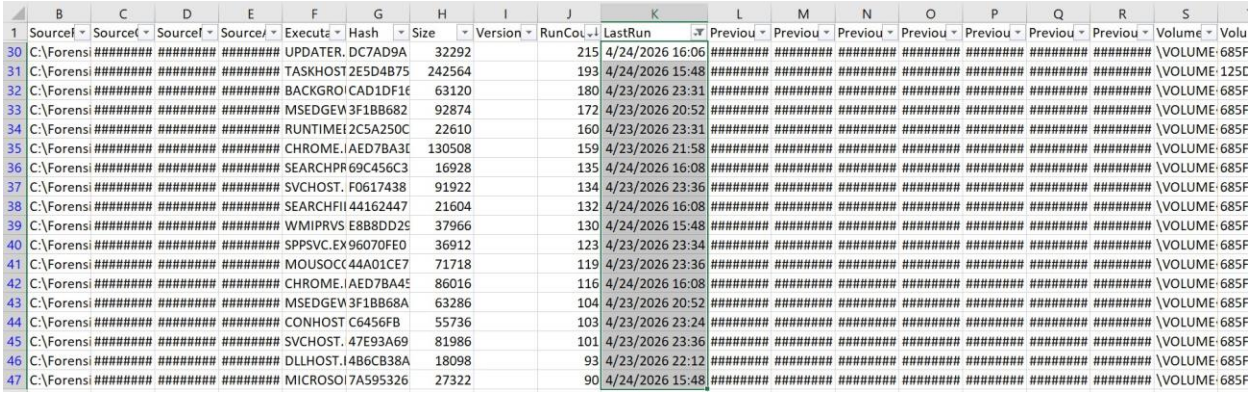
RecordNu	EventRecc	TimeCreat	EventId	Level	Provider	Channel	ProcessId	ThreadId	Computer	ChunkNur	UserId	MapDescr	UserName	RemoteHc	PayloadDa	PayloadDa	PayloadDa	PayloadDa	Pa
57097	57097	24:03.4	5379	LogAlways	Microsoft-	Security	1044	6764	Maryam	0		Credential	mrymm		SID: S-1-5-	Domain: N	LogonID: (	CountOfCred	
57098	57098	24:03.4	5379	LogAlways	Microsoft-	Security	1044	6764	Maryam	0		Credential	mrymm		SID: S-1-5-	Domain: N	LogonID: (	CountOfCred	
57099	57099	24:03.5	5379	LogAlways	Microsoft-	Security	1044	6764	Maryam	0		Credential	mrymm		SID: S-1-5-	Domain: N	LogonID: (	CountOfCred	
57100	57100	24:03.9	5379	LogAlways	Microsoft-	Security	1044	6764	Maryam	0		Credential	mrymm		SID: S-1-5-	Domain: N	LogonID: (	CountOfCred	
57101	57101	24:03.9	5379	LogAlways	Microsoft-	Security	1044	6764	Maryam	0		Credential	mrymm		SID: S-1-5-	Domain: N	LogonID: (	CountOfCred	
57102	57102	24:03.9	5379	LogAlways	Microsoft-	Security	1044	6764	Maryam	0		Credential	mrymm		SID: S-1-5-	Domain: N	LogonID: (	CountOfCred	
57103	57103	24:04.1	5379	LogAlways	Microsoft-	Security	1044	6764	Maryam	0		Credential	mrymm		SID: S-1-5-	Domain: N	LogonID: (	CountOfCred	
57104	57104	24:05.1	5379	LogAlways	Microsoft-	Security	1044	6764	Maryam	0		Credential	mrymm		SID: S-1-5-	Domain: N	LogonID: (	CountOfCred	
57105	57105	24:05.1	5379	LogAlways	Microsoft-	Security	1044	5416	Maryam	0		Credential	mrymm		SID: S-1-5-	Domain: N	LogonID: (	CountOfCred	
57106	57106	24:05.1	5379	LogAlways	Microsoft-	Security	1044	6764	Maryam	0		Credential	mrymm		SID: S-1-5-	Domain: N	LogonID: (	CountOfCred	
57107	57107	24:05.1	5379	LogAlways	Microsoft-	Security	1044	5416	Maryam	0		Credential	mrymm		SID: S-1-5-	Domain: N	LogonID: (	CountOfCred	
57108	57108	24:05.1	5379	LogAlways	Microsoft-	Security	1044	6764	Maryam	0		Credential	mrymm		SID: S-1-5-	Domain: N	LogonID: (	CountOfCred	
57109	57109	24:05.2	5379	LogAlways	Microsoft-	Security	1044	5416	Maryam	0		Credential	mrymm		SID: S-1-5-	Domain: N	LogonID: (	CountOfCred	
57110	57110	24:05.2	5379	LogAlways	Microsoft-	Security	1044	5416	Maryam	0		Credential	mrymm		SID: S-1-5-	Domain: N	LogonID: (	CountOfCred	
57111	57111	24:05.2	5379	LogAlways	Microsoft-	Security	1044	5416	Maryam	0		Credential	mrymm		SID: S-1-5-	Domain: N	LogonID: (	CountOfCred	
57112	57112	24:05.2	5379	LogAlways	Microsoft-	Security	1044	5416	Maryam	0		Credential	mrymm		SID: S-1-5-	Domain: N	LogonID: (	CountOfCred	
57113	57113	24:05.2	5379	LogAlways	Microsoft-	Security	1044	5416	Maryam	0		Credential	mrymm		SID: S-1-5-	Domain: N	LogonID: (	CountOfCred	
57114	57114	24:06.2	5379	LogAlways	Microsoft-	Security	1044	5416	Maryam	0		Credential	mrymm		SID: S-1-5-	Domain: N	LogonID: (	CountOfCred	

4. Part B: Execution Evidence (Prefetch Analysis)

Prefetch files were extracted from C:\Windows\Prefetch and parsed using PECmd. The dataset consisted of 312 .pf files, providing a highly detailed execution history.

4.1. Execution Inventory & Path Analysis

- **Primary Applications:** Run counts confirm the system is heavily used for development and browsing. Top executed binaries include CHROME.EXE, MSEDGEWEBVIEW2.EXE, and VS Code (CODE.EXE).
- **Suspicious Path Review:** Executions were identified originating from AppData\Local\Temp\ (e.g., _SETUP64.TMP, UNINS000.EXE). Analysis confirms these are legitimate software installers and temporary Python/pip package extractions, not malicious payloads.
- **External Media:** Zero executions were detected from USB or removable drives.



Sourcef	Sourcef	Sourcef	Sourcef	Executa	Hash	Size	Version	RunCou	LastRun	Previou	Previou	Previou	Previou	Previou	Previou	Previou	Volume	Volu
C:\Forensi	########	########	########	UPDATER.	DC7AD9A	32292		215	4/24/2026 16:06	########	########	########	########	########	########	########	\VOLUME	685F
C:\Forensi	########	########	########	TASKHOST	2E5D4B75	242564		193	4/24/2026 15:48	########	########	########	########	########	########	########	\VOLUME	125D
C:\Forensi	########	########	########	BACKGRO	CAD1DF1E	63120		180	4/23/2026 23:31	########	########	########	########	########	########	########	\VOLUME	685F
C:\Forensi	########	########	########	MSEDGEW	3F1BB682	92874		172	4/23/2026 20:52	########	########	########	########	########	########	########	\VOLUME	685F
C:\Forensi	########	########	########	RUNTIME	2C5A250C	22610		160	4/23/2026 23:31	########	########	########	########	########	########	########	\VOLUME	685F
C:\Forensi	########	########	########	CHROME.I	AED7BA3D	130508		159	4/23/2026 21:58	########	########	########	########	########	########	########	\VOLUME	685F
C:\Forensi	########	########	########	SEARCHPR	69C456C3	16928		135	4/24/2026 16:08	########	########	########	########	########	########	########	\VOLUME	685F
C:\Forensi	########	########	########	SVCHOST.	F0617438	91922		134	4/23/2026 23:36	########	########	########	########	########	########	########	\VOLUME	685F
C:\Forensi	########	########	########	SEARCHFI	44162447	21604		132	4/24/2026 16:08	########	########	########	########	########	########	########	\VOLUME	685F
C:\Forensi	########	########	########	WMIPRVS	E8B8DD29	37966		130	4/24/2026 15:48	########	########	########	########	########	########	########	\VOLUME	685F
C:\Forensi	########	########	########	SPPSVC.EX	96070FE0	36912		123	4/23/2026 23:34	########	########	########	########	########	########	########	\VOLUME	685F
C:\Forensi	########	########	########	MOUSOCC	44A01CE7	71718		119	4/23/2026 23:36	########	########	########	########	########	########	########	\VOLUME	685F
C:\Forensi	########	########	########	CHROME.I	AED7BA45	86016		116	4/24/2026 16:08	########	########	########	########	########	########	########	\VOLUME	685F
C:\Forensi	########	########	########	MSEDGEW	3F1BB68A	63286		104	4/23/2026 20:52	########	########	########	########	########	########	########	\VOLUME	685F
C:\Forensi	########	########	########	CONHOST	C6456FB	55736		103	4/23/2026 23:24	########	########	########	########	########	########	########	\VOLUME	685F
C:\Forensi	########	########	########	SVCHOST.	47E93A69	81986		101	4/23/2026 23:36	########	########	########	########	########	########	########	\VOLUME	685F
C:\Forensi	########	########	########	DLLHOST.I	4B6CB38A	18098		93	4/23/2026 22:12	########	########	########	########	########	########	########	\VOLUME	685F
C:\Forensi	########	########	########	MICROSO	7A595326	27322		90	4/24/2026 15:48	########	########	########	########	########	########	########	\VOLUME	685F

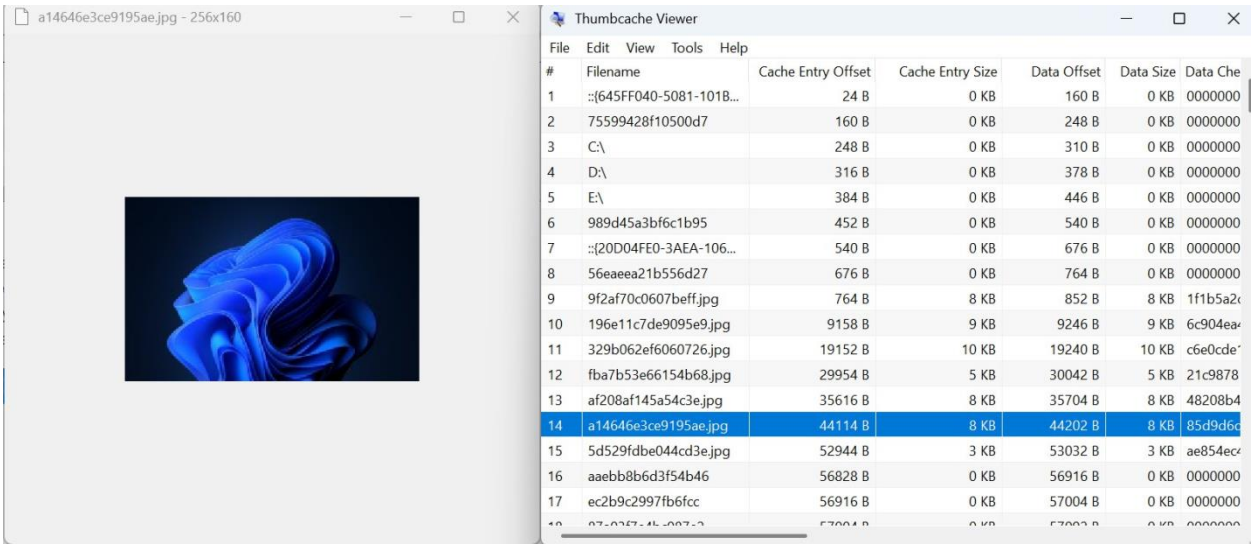
5. Part C: User Activity & Deletion Artifacts

This phase focused on persistent traces of visual media and deleted content.

5.1. Cache & Thumbcache

- **Thumbcache:** Recovered 15 databases from AppData\Local\Microsoft\Windows\Explorer, ranging from thumbcache_16.db to thumbcache_1280.db. Using Thumbcache Viewer, cached thumbnails (including a Windows wallpaper image) were verified, proving visual interaction with local media.

- **Browser Cache:** Extracted Google Chrome cache (Profile 2). Metadata confirms routine web browsing sessions, with no immediate connections to known malicious infrastructure.



5.2. Recycle Bin Analysis

Analysis of the C:\$Recycle.Bin directory for the active user SID (S-1-5-21-...-1001) yielded significant deleted file artifacts.

- **Finding:** Successfully recovered both metadata ($I) and content ($R) files.
- **Deleted File Types:** Evidence confirms the recent deletion of executables (.exe), archives (.zip), documents (.docx), and images (.png).

```
C:\$Recycle.Bin>dir /a
 Volume in drive C has no label.
 Volume Serial Number is 685F-3A4B

 Directory of C:\$Recycle.Bin

04/17/2026  10:16 AM    <DIR>          .
04/23/2026  01:38 PM    <DIR>          ..
04/17/2026  07:17 AM    <DIR>          S-1-5-21-256890474-1904052976-3891447156-1000
04/23/2026  03:01 PM    <DIR>          S-1-5-21-256890474-1904052976-3891447156-1001
               0 File(s)              0 bytes
               4 Dir(s)  301,538,172,928 bytes free
```

```
 Directory of C:\$Recycle.Bin\S-1-5-21-256890474-1904052976-3891447156-1001

04/23/2026  03:01 PM    <DIR>          .
04/17/2026  10:16 AM    <DIR>          ..
04/17/2026  11:51 AM               114 $I08LSQ4.exe
04/17/2026  11:17 AM               112 $I1PKX66.lnk
04/19/2026  12:37 PM               104 $I26XRLR
04/22/2026  09:36 PM               158 $I55RZ0A.zip
04/22/2026  09:36 PM               158 $IFE4Z58.zip
04/22/2026  09:36 PM               150 $IG9PKOV.zip
04/17/2026  11:06 AM               118 $IH250KO.lnk
04/19/2026  12:33 PM               120 $IHXRO6X.lnk
04/17/2026  11:51 AM               144 $IID7H20.exe
04/23/2026  03:01 PM                72 $IL0GV9O
04/17/2026  02:59 PM               136 $IL9RHTE.docx
04/17/2026  11:06 AM               110 $IPXZSQF.lnk
04/19/2026  11:44 AM               114 $IRJRWQE.lnk
04/23/2026  03:01 PM                70 $ISHX4UE
04/17/2026  11:46 AM               114 $IVEF0QU.exe
04/17/2026  11:16 AM               100 $IVNDRIZ.png
04/19/2026  12:37 PM               100 $IWDDGZV
04/23/2026  02:56 PM                94 $IXACHF9
04/23/2026  02:56 PM                98 $IYE22AA
04/17/2026  11:50 AM         6,876,208 $R08LSQ4.exe
04/17/2026  11:03 AM             2,206 $R1PKX66.lnk
04/19/2026  12:25 PM    <DIR>          $R26XRLR
04/19/2026  01:27 PM            17,619 $R55RZ0A.zip
04/19/2026  01:27 PM            17,619 $RFE4Z58.zip
04/19/2026  01:26 PM            17,619 $RG9PKOV.zip
04/17/2026  11:04 AM             2,206 $RH250KO.lnk
04/19/2026  12:19 PM               839 $RHXRO6X.lnk
04/17/2026  11:41 AM       177,448,040 $RID7H20.exe
04/23/2026  02:56 PM    <DIR>          $RL0GV9O
04/17/2026  02:59 PM            22,895 $RL9RHTE.docx
04/17/2026  11:04 AM             2,206 $RPXZSQF.lnk
04/19/2026  11:31 AM             2,356 $RRJRWQE.lnk
04/23/2026  02:56 PM    <DIR>          $RSHX4UE
04/17/2026  11:46 AM         6,876,208 $RVEF0QU.exe
```

6. Cross-Artifact Correlation

The integrity of the investigation is validated by cross-referencing the three artifact categories:

1. **Authentication to Execution:** Interactive logons (Event ID 4624) directly precede the execution timestamps of VS Code and Chrome (Prefetch timeline).
2. **Execution to File Interaction:** Execution of development environments and browsers correlates with the creation of Python temp files, Chrome cache generation, and the eventual deletion of project files (Recycle Bin $I/$R records).

The overlapping timelines present a unified, consistent narrative of standard administrative and development usage with zero conflicting anomalies.